

Interactive Map of Direct Flight Times & Distances

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DirectFlightTimes.Streamlit.app

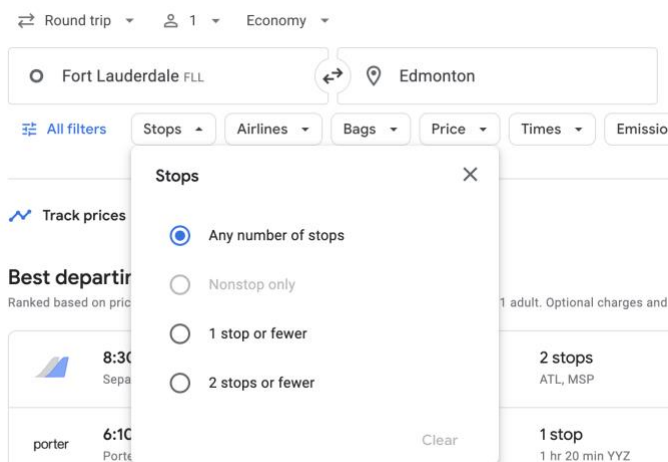
Abstract

My project aims to create an interactive web application that allows users to quickly calculate the direct flight duration between two selected cities. The end-product is an interactive map overlaid with markers of the large airports across the world. Users will select two cities and the application produces the distance as well as the estimated direct flight time. For this project, I leveraged Streamlit for the frontend interface, Folium for the map visualization, and Geopy for distance calculations, providing simple and fun tool for prospective travelers and geographic education.

Introduction

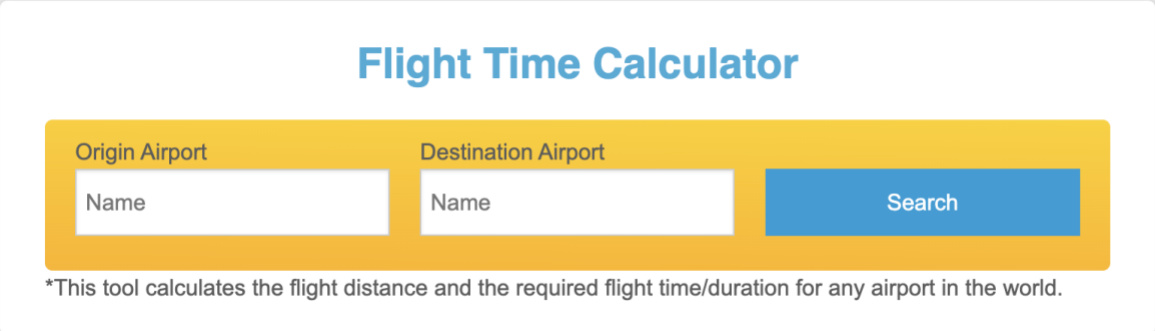
Motivation & Background

According to Forbes, 40% of Americans are planning to travel more in 2024 than 2023, driven by Millennials and Gen Z populations [1]. With this increased interest in traveling, there is a need for quick and easy interactive tools that help users discover the spatial relationships and travel times between cities across the world. This project addresses this need by offering a simple, user-friendly application that provides a flight path and estimated flight durations between cities. I believe a tool like this can be useful for prospective travelers, and inquisitive geography enthusiasts. Most recently, I was curious about the direct flight time from Fort Lauderdale, Florida (FLL) to Edmonton, Alberta (YEG). The problem is that there are currently no commercial airlines that offer direct flights from FLL to YEG so I would be unable to find that information readily on say, a Google Flights search.



Google Flights showing no "nonstop only" flights from FLL to YEG

There are existing tools I have found online such as Flight Time Calculator [2], Sky Vector [3], or Airplane Manager [4]. Flight Time Calculator was my main inspiration for this project as it's a tool that "...calculates the flight distance and the required flight time/duration for any airport in the world". You simply type the origin airport, destination airport and it computes the estimated distance and flight time. Flight Time Calculator however does not include a map with your selected flight path between the two airports that you chose.



Flight Time Calculator

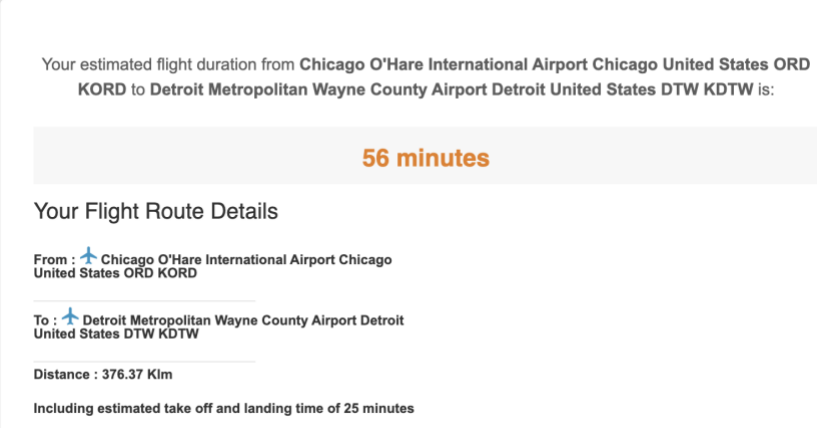
Origin Airport
Name

Destination Airport
Name

Search

*This tool calculates the flight distance and the required flight time/duration for any airport in the world.

<https://www.flighttimecalculator.org/>



Your estimated flight duration from **Chicago O'Hare International Airport Chicago United States ORD** to **Detroit Metropolitan Wayne County Airport Detroit United States DTW KDTW** is:

56 minutes

Your Flight Route Details

From :  **Chicago O'Hare International Airport Chicago United States ORD KORD**

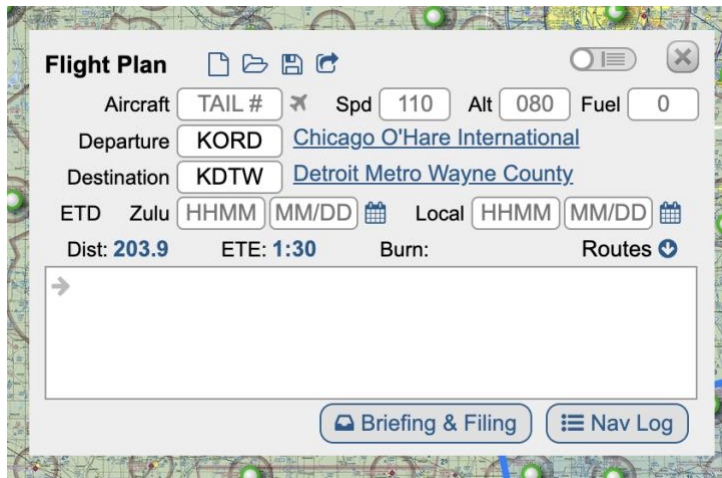
To :  **Detroit Metropolitan Wayne County Airport Detroit United States DTW KDTW**

Distance : 376.37 Km

Including estimated take off and landing time of 25 minutes

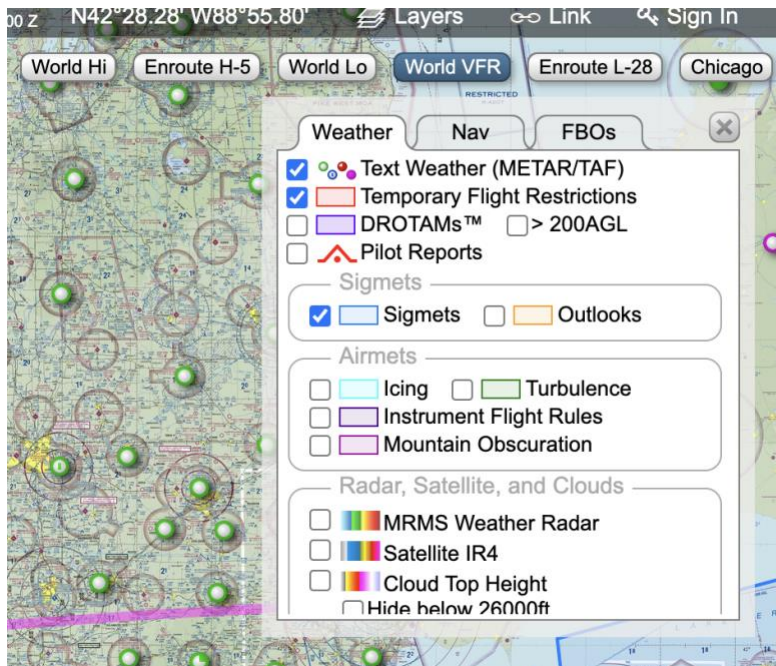
<https://www.flighttimecalculator.org/>

Sky Vector is one tool I found that allows users to click on airport markers on a map and get an estimated flight time and distance. However, you must change the speed in the flight plan box, or it defaults to a value of 110, which I assume is miles per hour. This requirement can be a barrier for those seeking a simple tool to provide approximate commercial flight times without prior knowledge of their aircraft or its relative speed.



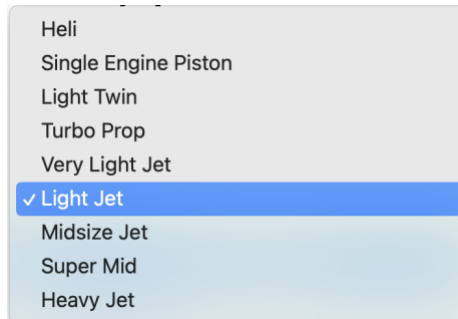
<https://skyvector.com/>

The map is overlaid with airports but includes a level of detail that may not interest the average traveler. For example, there are several layers to choose from, and once activated, it is unclear how to revert to the main map view you started with. Additionally, the numerous options and their acronyms can be confusing. Overall, I believe this tool is geared more towards pilots or aeronautic enthusiasts who have more knowledge in the aviation industry.



<https://skyvector.com/>

The flight calculator from Airplane Manager is an effective tool with the ability to search two airports and get the estimated flight time with the addition of a flight map. It gives you options to adjust the seasonal wind patterns (seasonal winds vs no winds) as well as the size of the airplane you would be flying in.



<https://airplanemanager.com/flight-calculator>

In addition to the map displaying flight path, distance, and flight time, Flight Manager also includes details such as altitude, cruise speed, and heading. The layout is simple, ensuring you are not inundated with information, unlike Sky Vector. What sets my project apart is the ability for users to browse the map, click on two airport markers, and automatically view the estimated distance and flight path. Drawing inspiration from the three mentioned websites, my project aims to create a more simplistic approach to finding the estimated flight time from airport A to airport B.

Flight Calculator

KJFK	KLAX
Seasonal Winds	▼
Light Jet	▼
CALCULATE	↔ REVERSE
Time Difference	-3
Flight Time	5 hours 37 mins
Distance	2146 NM
Altitude	39000 FT
Heading	274 °
Cruise Speed	410 kts

KJFK - John F Kennedy Intl	
Runway Length	14511 ft
Elevation	13 ft

KLAX - Los Angeles Intl	
Runway Length	12091 ft
Elevation	125 ft

A map of the United States showing a red curved line representing the flight path from Los Angeles, California to New York, New York. The map includes labels for major cities like Los Angeles, Phoenix, Washington, New York, and La Habana. The flight path starts near Los Angeles and ends near New York.

<https://airplanemanager.com/flight-calculator>

Objectives

Interactive Map: To create a map where users can click on city markers to select their departure and destination cities.

Distance and Duration Calculation: To calculate and display the direct flight distance and estimated duration between the selected cities.

User Engagement: To provide an engaging and fun experience through an interactive visualization.

Methods

Streamlit: Streamlit is a free and open-source framework that provides the ability to deploy web applications in Python without writing any of the front-end code. I utilized Streamlit to launch my interactive web application interface, allowing for users to select airport markers, displaying the results dynamically.

Folium: Folium is a data visualization Python library specifically for geospatial data and mapping. For this project, Folium is used to place city markers and leverage its interactive map functionalities.

Geopy: The Python library Geopy to extract the geodesic distance between two points. For this project this is the distance between two cities that represent a direct flight distance.

Interactive Markers: The map features markers for each airport's coordinate location, providing users with a view of all available airports worldwide. Once the map refreshes with the flight path, it displays only the origin and destination airports. I have implemented plane-departure and plane-arrival icons to provide helpful visual cues for users.

Distance and Duration Display: After selecting two cities, the application will display the calculated flight distance and estimated duration.

Process

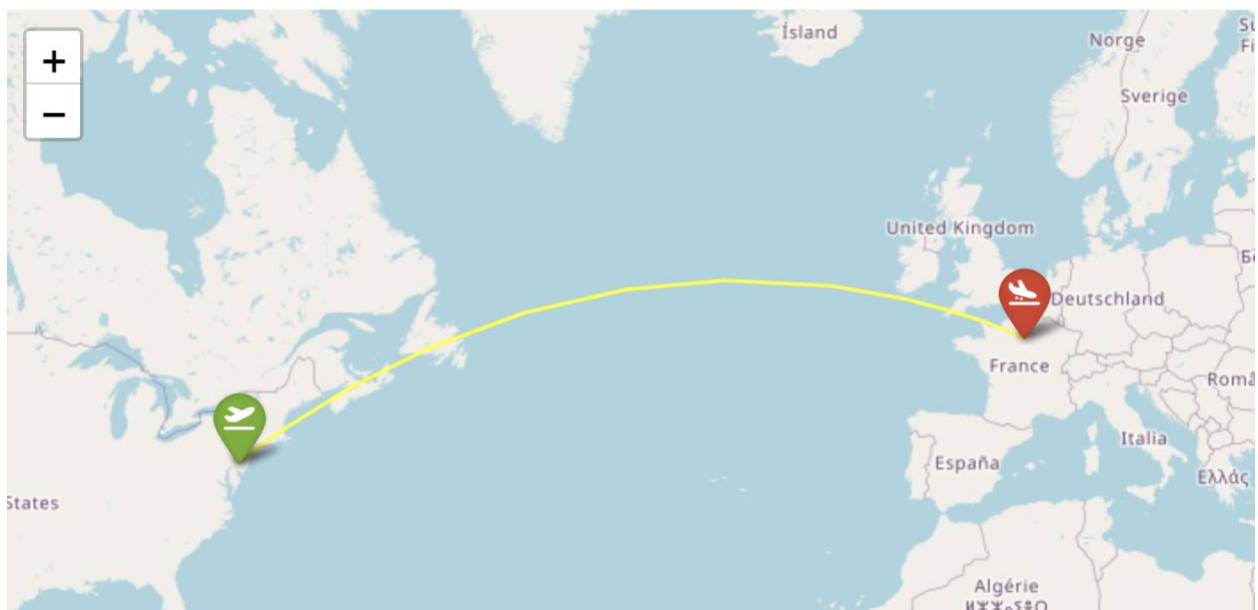
The dataset used was from the Bureau of Transportation Statistics (BTS) which includes the state, country and their respective coordinate information for all relevant airports. The BTS dataset is highly suitable for this project because it contains precise geographic coordinates necessary for distance calculations. I did however clean and purge unnecessary information to make sure there were no unforeseen errors. The simplicity of this dataset ensures there will be quick and accurate processing.

I started with the master BTS download [6] of all airport listings and filtered it by removing closed airports and duplicate entries. To avoid crowding the map with tens of thousands of

small & non-commercial airports worldwide, I limited the number of airports to only "large airports" as designated by the dataset from <https://ourairports.com/world.html>. This dataset includes much of the same information as the BTS file, with the addition of the "type" classification for small, medium, and large airports. I cross-referenced the airport codes between the two files and removed any entries that were not classified as large airports. After this reduction, my file contained 455 airport locations.

I experienced several failed experiments during the development of this application, most of which involved ensuring the website's proper functionality. I began by loading the data and projecting the map and airport location markers using the Folium library. Next, I implemented the functionality to distinguish between "valid clicks" on the map (i.e. airport markers) and "invalid clicks", from clicking and dragging on the map. This step took the longest to perfect as it was challenging to figure out the correct logic that worked with the Streamlit session state.

One functionality that I was unable to deploy was replacing the Folium map (Mercator) with a 3D globe view to provide a more accurate representation of travel across vast distances. To reconcile the inability to implement a 3D map, I calculated and projected the great circle path with Geodesic, from city A to city B. I found this to be an improvement over simply projecting a straight line from city A to city B.



Flight path from Philadelphia, PA to Paris, France

When I first set up this application, I used a speed placeholder of 500 miles per hour (mph). Afterwards, I did some research on flight speeds and found a lot of conflicting information online. I gathered that commercial airline speeds range from 400 to 700 mph, with longer-range airplanes generally having higher cruising speeds. I also found that short-haul flights

are considered to be less than 700 miles, long-haul flights are greater than 3,000 miles, and medium-haul flights fall in between.

Using this information, I decided to base the speed of an airplane on the flight distance. I chose 400 mph for short-haul flights (<700 miles), 500 mph for medium-haul flights (700-3,000 miles), and 550 mph for long-haul flights (>3,000 miles). Although these are the speeds I decided on, I plan to revisit this metric in the future to provide a more accurate calculation.

Flight Details

Trip Type: Short Haul

Origin: Miami, FL (Metropolitan Area)
(Miami International, MIA)

Destination: New Orleans, LA (Louis
Armstrong New Orleans International, MSY)

Distance between: 674 miles

Estimated flight duration: 1 hours 41
minutes

Flight Details

Trip Type: Medium Haul

Origin: Chicago, IL (Chicago O'Hare
International, ORD)

Destination: New York City, NY
(Metropolitan Area) (John F. Kennedy
International, JFK)

Distance between: 740 miles

Estimated flight duration: 1 hours 28
minutes

Flight Details

Trip Type: Long Haul

Origin: Detroit, MI (Detroit Metro Wayne
County, DTW)

Destination: Tel Aviv, Israel (Ben Gurion,
TLV)

Distance between: 5997 miles

Estimated flight duration: 10 hours 54
minutes

Results

Below, you will see some examples and a walk through of the steps of my website.

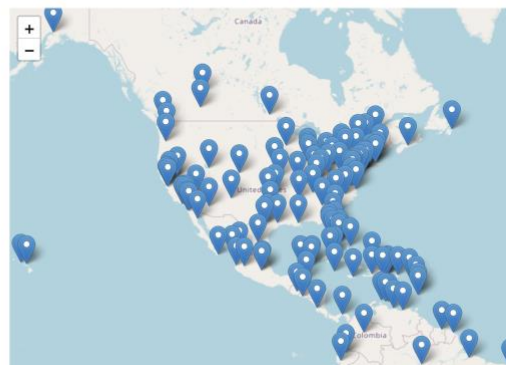
Landing page

Flight Details

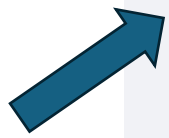
Interactive Flight Duration Map

Click on two airports on the map to calculate the direct flight duration

Reset Selections



Select an airport and the screen will refresh, reflecting your chosen airport in the “Flight Details” sidebar.

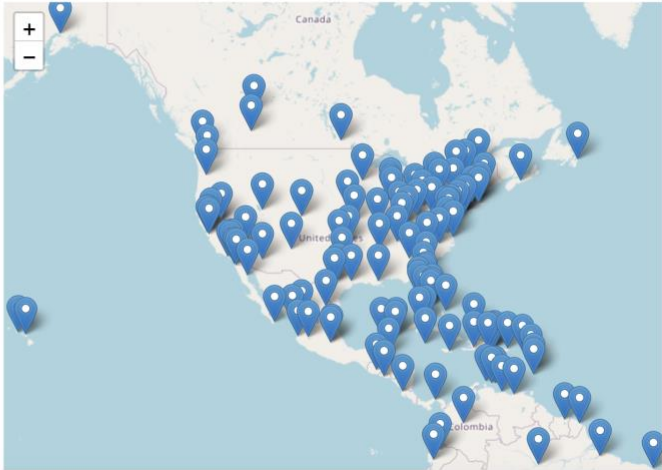


Flight Details
Origin: San Francisco, CA (Metropolitan Area)
(San Francisco International, SFO)

Interactive Flight Duration Map

Double-click on two airports on the map to calculate the direct flight duration

Reset Selections



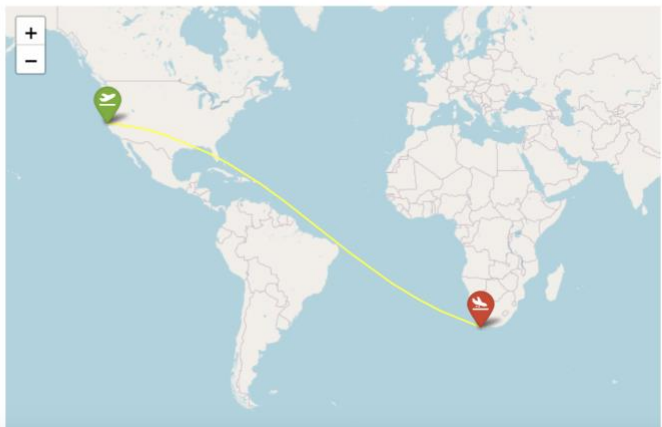
Once you’ve selected the second airport, the screen refreshes again, displays the complete trip details and hides all other airport markers. The plane-depart and plane-arrival icons were imported using Font Awesome.

Flight Details
Trip Type: Long Haul
Origin: San Francisco, CA (Metropolitan Area)
(San Francisco International, SFO)
Destination: Cape Town, South Africa (Cape Town International, CPT)
Distance between: 10252 miles
Estimated flight duration: 18 hours 38 minutes

Interactive Flight Duration Map

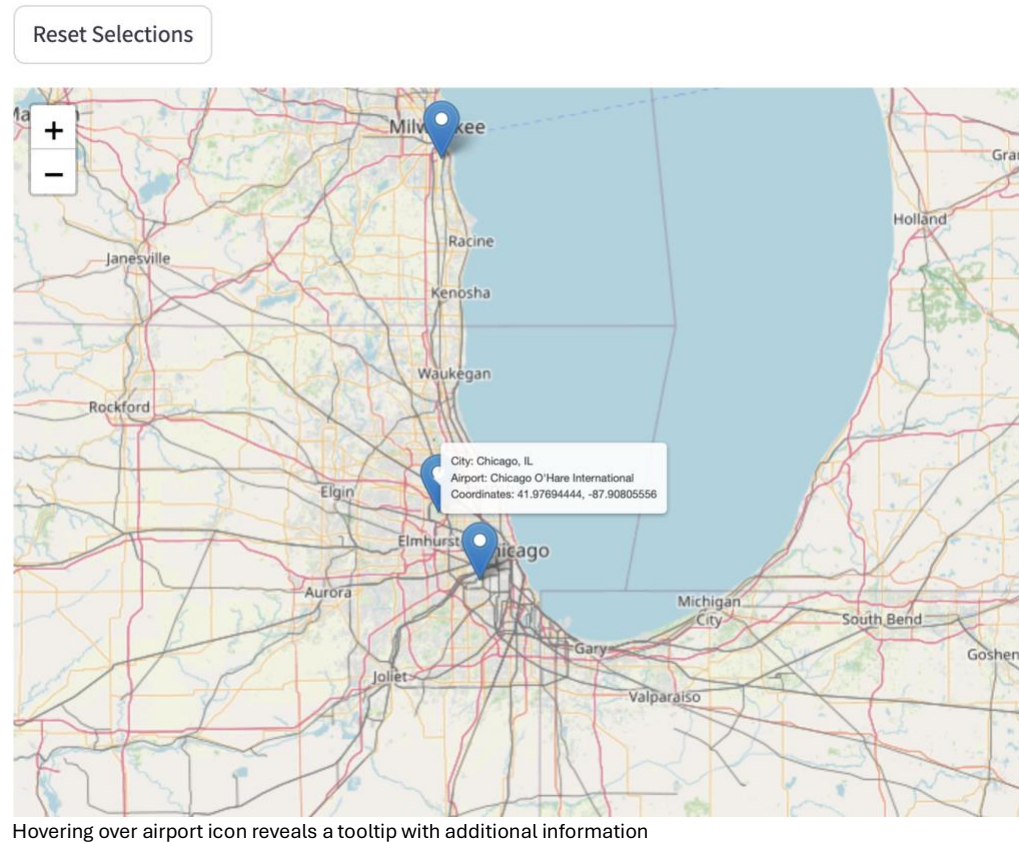
Double-click on two airports on the map to calculate the direct flight duration

Reset Selections



If you wish to start over and reselect a pair of airports, simply click “Reset Selections” above the map. The screen will refresh, and you will see all airport icons once again.

A tooltip was also added with details such as city, airport name and coordinates of the airport you hover over.



Hovering over airport icon reveals a tooltip with additional information

As far as insights go, this project did not provide much due to its design as a fun tool for users to explore different direct flight times and distances. However, I believe that providing the flight path as a great-circle path visualization helps users understand why flights often appear to take curved routes on flat maps, which is educational in itself.

Conclusion:

My flight tool provides a fun, interactive web application that allows users to quickly calculate the estimated direct flight duration and distance between two selected cities. This was achieved using Streamlit for the frontend interface, Folium for the map visualization, and Geopy for distance calculations. While the end result meets my project goals, I realize there is still room for improvement.

One enhancement I would make is to start by downloading all BTS flight records from the previous year, cleaning the file, and extracting each pairwise origin and destination route that is currently flown. To my knowledge, this information exists for all foreign and domestic flights, as long as either the origin or destination is in a U.S. state or territory. I would add functionality to my tool to inform users if the direct flight option they selected currently

exists and, if so, which airline carrier offers that flight. Moreover, I would like to include a search box feature for users who cannot locate their desired destination on the map. Lastly, as previously mentioned, it would be beneficial to add more dynamic logic to the flight speed information.

References

[1] Pokora, B. (2024, January 8). *2024 vs. 2023 Travel trends: 40% of Americans plan to travel more this year*. Forbes Advisor. <https://www.forbes.com/advisor/credit-cards/travel-rewards/survey-travel-plans-2024/>

[2] (n.d.). *Flight Time Calculator*. Flight Time Calculator. <https://www.flighttimecalculator.org/>

[3] (n.d.). *SkyVector: Flight Planning / Aeronautical Charts*. SkyVector. <https://skyvector.com/>

[4] (n.d.). *Airplane Manager*. AirplaneManager. <https://airplanemanager.com/>

[5] "Flight Length." Wikipedia, Wikimedia Foundation, 7 July 2024, [en.wikipedia.org/wiki/Flight_length#:~:text=United%20Airlines%20defines%20short%20haul,2%2C600%20nmi%3B%204%2C800%20km\)](https://en.wikipedia.org/wiki/Flight_length#:~:text=United%20Airlines%20defines%20short%20haul,2%2C600%20nmi%3B%204%2C800%20km).).

[6] Download page. (n.d.). https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FLL&QO_fu146_anzr=N8vn6v10%20f722146%20gnoyr5